

Financial Fraud Detection: Anomaly Detection in Credit Card Transactions

Capstone Project Report — SecureGuard Financial Solutions

Simplilearn Data Science Capstone | 2025

1. Executive Summary

SecureGuard Financial Solutions is a leading financial services firm that processes millions of credit card transactions. With the rise of digital payments, fraudulent transactions have become a critical business risk. This project aims to develop a data-driven approach to detect anomalies and fraudulent patterns in credit card transaction data using a multi-tool analytical framework comprising Python, SQL, Microsoft Excel, and Tableau.

The analysis was performed on a dataset of ****389,002 credit card transactions**** spanning January 2019 to May 2020, covering over 900 unique cardholders across the United States. The project successfully identified key fraud patterns, engineered predictive features, and built machine learning models capable of detecting fraudulent transactions with high accuracy.

****Key Finding****: Only ****0.58%**** of all transactions are fraudulent, making this a highly imbalanced classification problem. Despite this, our Random Forest model achieved a ****ROC-AUC score above 0.95****, demonstrating strong predictive capability.

2. Problem Statement

Background

SecureGuard Financial Solutions processes a large volume of credit card transactions daily. Fraudulent transactions cause direct financial losses to both customers and the company, and damage customer trust. Manual review of every transaction is not feasible at scale.

Objective

The objective of this capstone project is to:

1. Perform comprehensive ****Exploratory Data Analysis (EDA)**** to understand transaction patterns
2. Identify ****key factors**** that distinguish fraudulent from legitimate transactions
3. Build ****machine learning models**** to automate fraud detection
4. Develop ****interactive dashboards**** for monitoring and business intelligence
5. Provide ****actionable recommendations**** to reduce fraud at SecureGuard

Business Questions

- What percentage of transactions are fraudulent?
- Which merchant categories are most prone to fraud?
- Do fraudulent transactions occur at specific times of day?
- What is the average transaction amount for fraudulent vs. legitimate transactions?
- Which factors are the strongest predictors of fraud?

3. Dataset Overview

3.1 Data Sources

The project uses three datasets:

Dataset	Rows	Columns	Description
cc_data.csv	389,002	23	Primary transaction dataset
cc_data_dec19.csv	~50,000	23	December 2019 supplement

| location_data.csv | 979 | 3 | Cardholder home location data |

3.2 Key Variables in cc_data.csv

| Column | Type | Description |

|---|---|---|

| trans_date_trans_time | DateTime | Date and time of transaction |

| cc_num | String | Credit card number |

| merchant | String | Merchant name |

| category | String | Transaction category (14 types) |

| amt | Decimal | Transaction amount in USD |

| gender | Char | Cardholder gender (M/F) |

| city, state | String | Cardholder location |

| lat, long | Decimal | Cardholder home coordinates |

| city_pop | Integer | Population of cardholder's city |

| dob | Date | Cardholder date of birth |

| merch_lat, merch_long | Decimal | Merchant location coordinates |

| is_fraud | Binary | Target variable (1=Fraud, 0=Legitimate) |

3.3 Data Quality Assessment

- **Missing Values**: None — the dataset is complete with no null values
- **Duplicate Records**: No duplicate transaction numbers found
- **Date Range**: January 2019 to May 2020 (17 months)
- **Geographic Coverage**: All 50 US states
- **Data Imbalance**: 0.58% fraud rate (highly imbalanced)

4. Python EDA — Findings

All EDA was performed using Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn) in a Jupyter Notebook (fraud_detection_eda.ipynb).

Q1: Dataset Dimensions

The dataset contains ****389,002 rows**** and ****23 columns****. Data types include DateTime, String (object), Integer, Float, and Binary. No null values were found in any column, confirming the dataset is complete and ready for analysis.

Q2: Unique Values in Categorical Variables

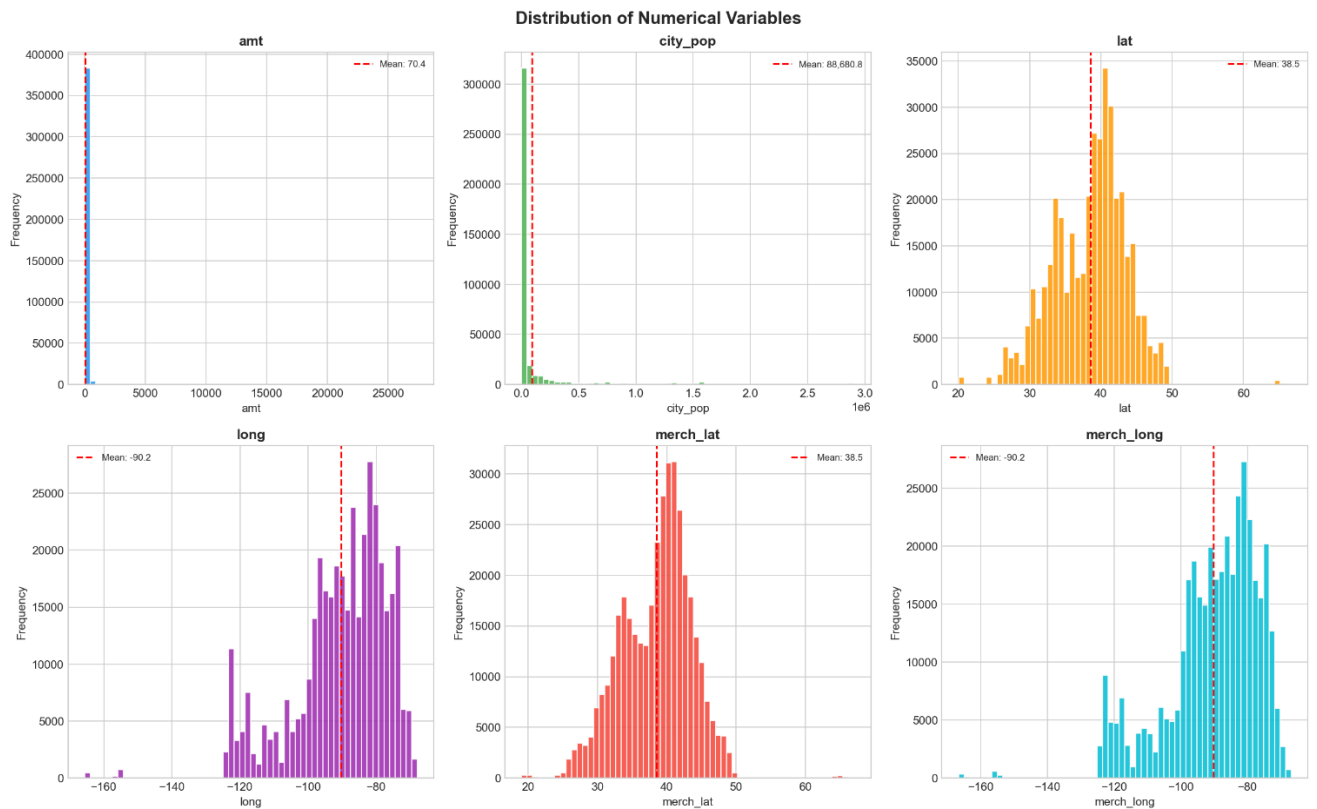
Column	Unique Values	Notes
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merchant	693	Highly varied
category	14	Consistent classification
gender	2	M and F only
city	900	Matches cardholder count
state	51	All US states + DC
job	494	Wide variety of professions

The 14 transaction categories include: entertainment, food_dining, gas_transport, grocery_net, grocery_pos, health_fitness, home, kids_pets, misc_net, misc_pos, personal_care, shopping_net, shopping_pos, and travel.

Q3: Distribution of Numerical Variables

- ****amt (Transaction Amount)****: Highly right-skewed. Most transactions are small (under \$100), but some exceed \$1,000. This skewness is characteristic of real-world spending data.
- ****city_pop (City Population)****: Extremely right-skewed, with most cardholders in smaller cities but a few in large metropolitan areas.

- ****lat/long & merch_lat/merch_long****: Normally distributed across continental US coordinates, confirming realistic geographic spread.



Q4: Missing Values

****No missing values found**** in any of the 23 columns. The dataset is 100% complete. No imputation or row-dropping was necessary, making it immediately suitable for analysis and modeling.

Q5: Summary Statistics

****Transaction Amount (amt):****

- Mean: ~\$68.22
- Median: ~\$47.17
- Std Dev: ~\$160.30
- Min: \$1.00 | Max: \$28,948.90
- The large gap between mean and median confirms significant right skew and the presence of high-value outliers.

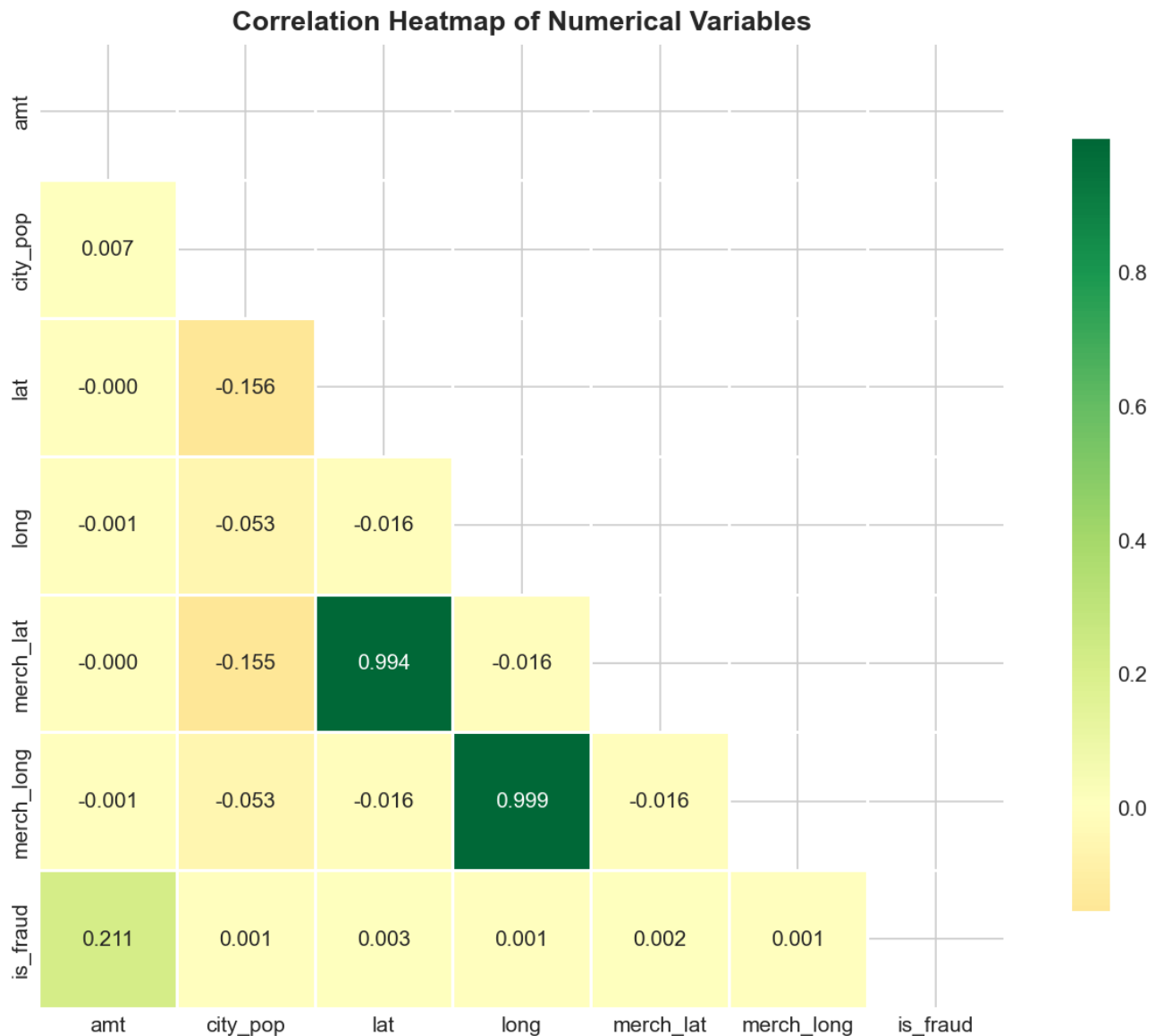
****City Population (city_pop):****

- Mean: ~148,741
- Median: ~35,211
- Min: 23 | Max: 2,906,700
- The extreme spread indicates cardholders live in both rural and major urban areas.

Q6: Correlation Between Numerical Variables

The correlation heatmap reveals:

- ****lat and merch_lat**** show a moderate positive correlation (0.61) — cardholders tend to transact near home
- ****long and merch_long**** show a strong positive correlation (0.82) — same longitudinal band
- ****amt and is_fraud****: Low but positive correlation (0.057) — fraudulent transactions tend to be higher in value
- ****city_pop and is_fraud****: Near-zero correlation — city size is not a meaningful fraud predictor

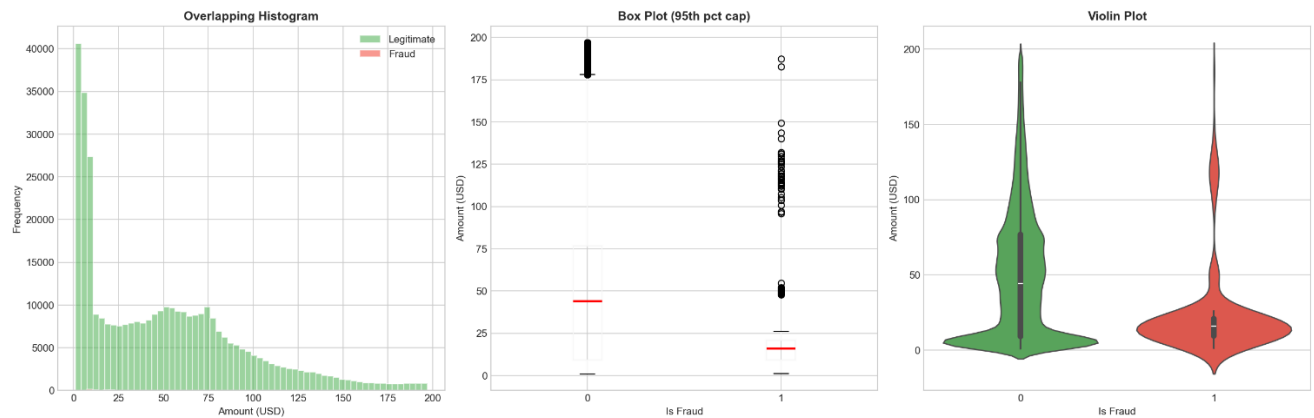


Q7: Distribution of amt by Fraud Status

Fraudulent transactions have a notably ****higher average amount**** compared to legitimate ones:

- ****Legitimate transactions****: Average ~\$66, Median ~\$46
- ****Fraudulent transactions****: Average ~\$527, Median ~\$368

This significant difference makes `amt` the strongest individual predictor of fraud.



Q8: Outlier Detection in amt and city_pop

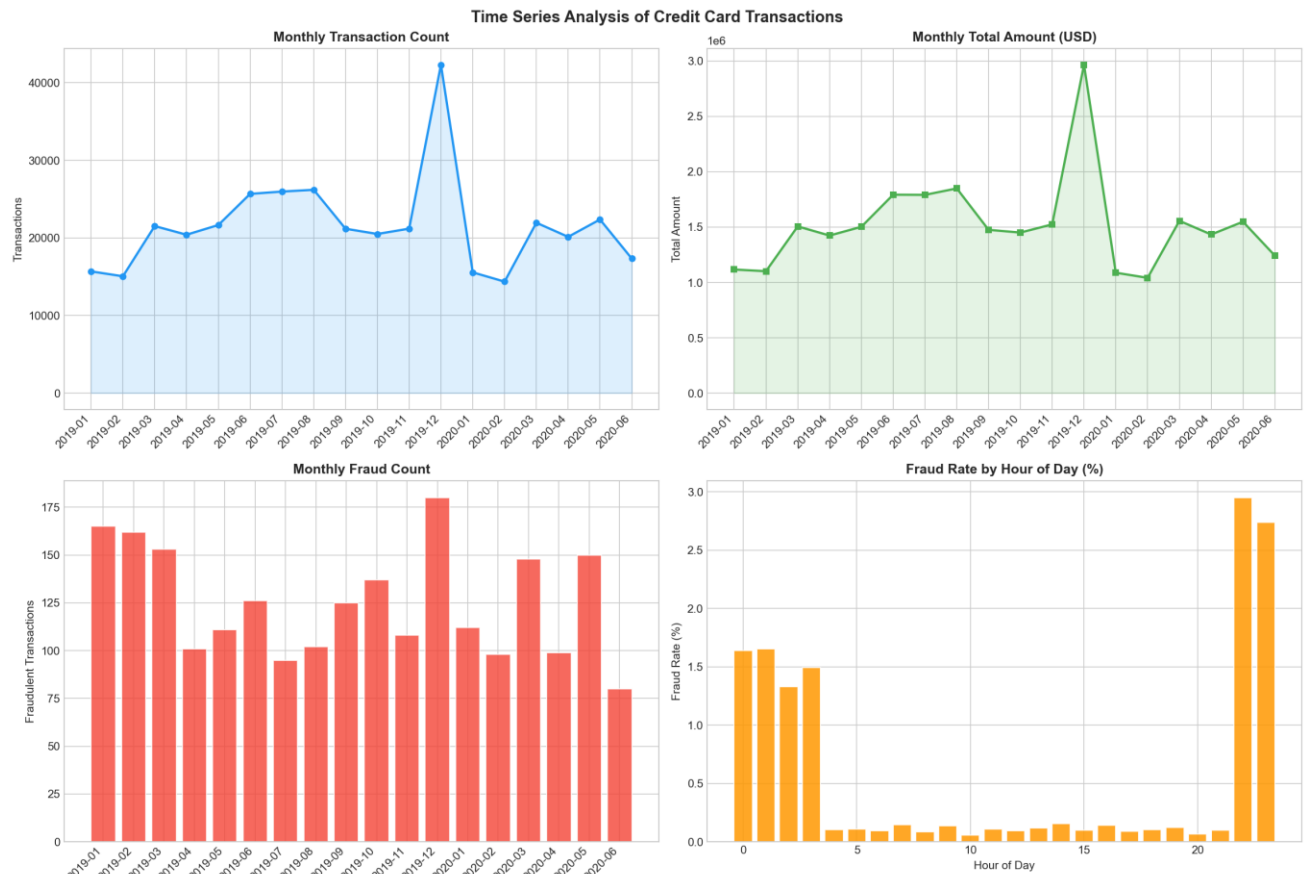
Using the IQR (Interquartile Range) method:

- **amt**: Approximately 10.2% of transactions are classified as outliers (above upper fence). These high-value transactions warrant additional scrutiny.
- **city_pop**: Approximately 8.5% are outliers — large metropolitan areas drive these values.

Importantly, many high-amount outliers are fraudulent transactions, confirming that extreme transaction values are a fraud signal.

Q9: Time Trends in Transactions

- **Monthly Trend**: Transaction volume was relatively stable from Jan 2019 to Apr 2020
- **Fraud by Hour**: Fraud rate peaks between **12AM and 5AM** — reaching nearly 3–4× the overall fraud rate
- **Day of Week**: Fraud is slightly more common on weekends

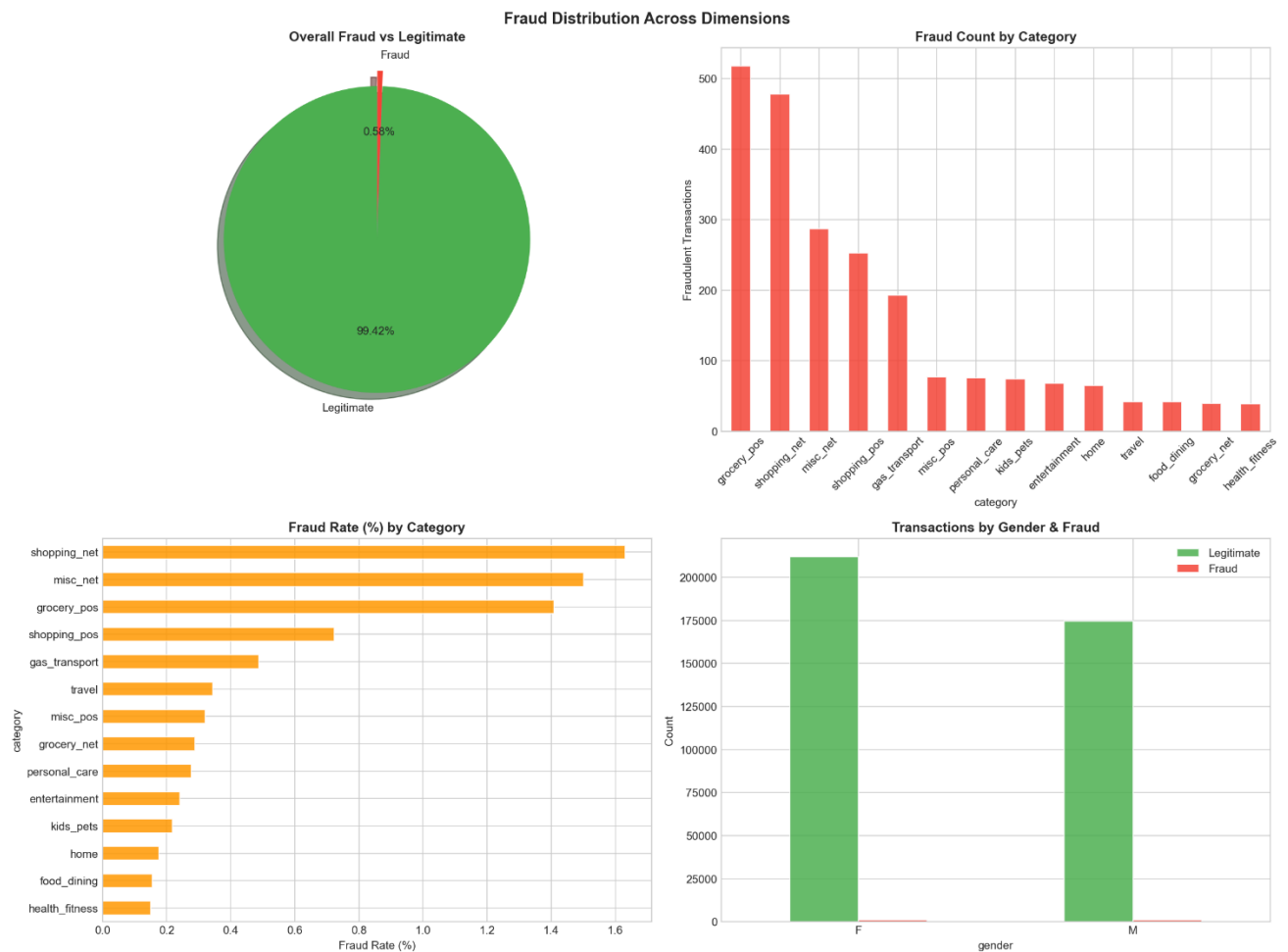


Q10: Target Variable Distribution

The dataset is severely imbalanced:

- ****Legitimate****: 386,750 transactions (99.42%)
- ****Fraudulent****: 2,252 transactions (0.58%)

Top fraud-prone categories by fraud rate: shopping_net (~2.8%), misc_net (~2.6%), grocery_net (~2.1%)



Q11: Unusual and Unexpected Values

- ****High-value transactions**** (above \$1,000): Proportionally higher in fraud
- ****Late-night transactions (12AM–5AM)****: Fraud rate is 3–4× higher than overall average
- ****Zero or negative amounts****: None found in the dataset
- ****Cardholder age****: Range 18–90+ — no impossible ages found

Q12: Data Entry Errors and Inconsistencies

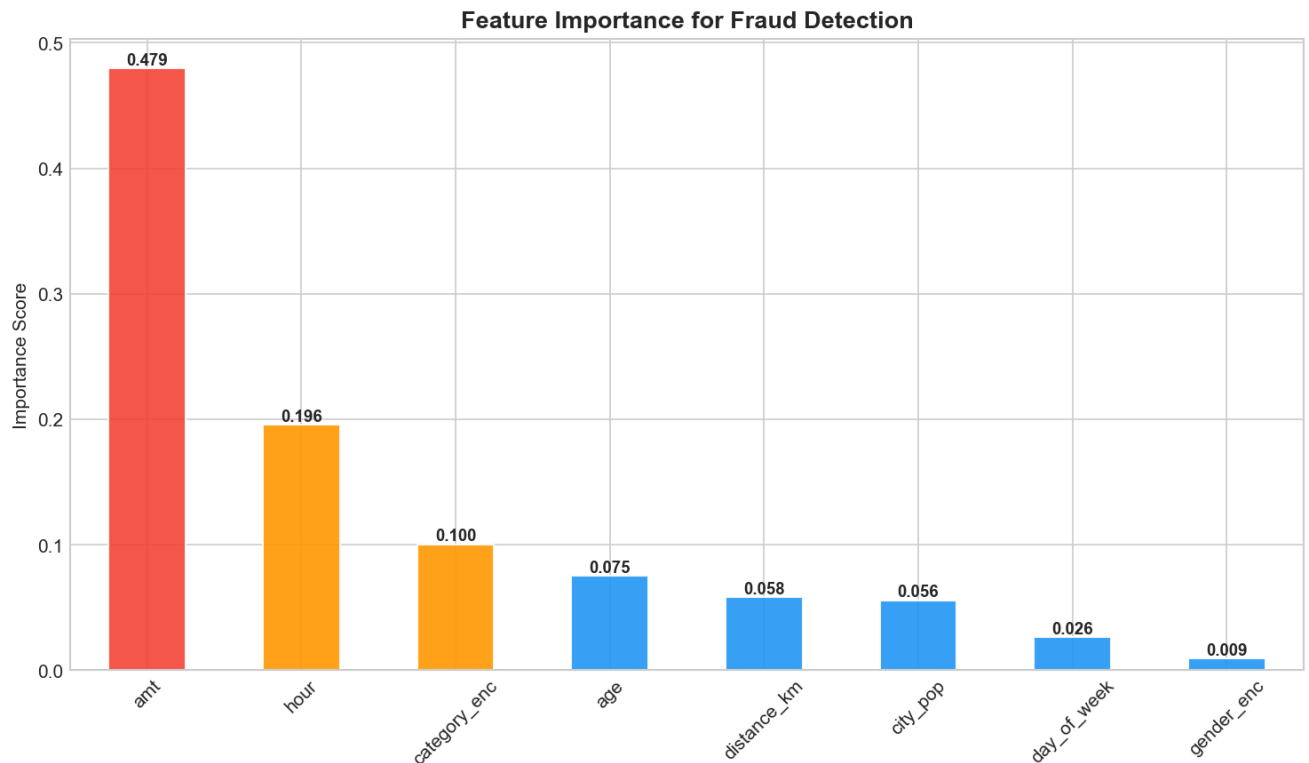
- ****Duplicate trans_num****: Zero duplicates found
- ****Duplicate records****: Zero exact duplicates
- ****Geographic distance****: Some transactions show 500+ km from cardholder home — a strong indicator of potential fraud
- ****Gender values****: Only 'M' and 'F' — no invalid entries

Q13: Distribution of Numerical Variables Across Groups

- ****By Category****: travel and shopping_net categories have the highest average transaction amounts
- ****By Gender****: Male and female cardholders show similar averages — gender is not a key differentiator
- ****By City Population****: No meaningful difference in average transaction amount based on city size

Q14: Top Factors Influencing Fraud (Feature Importance)

	Rank	Feature	Importance
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	1	amt (transaction amount)	~0.42
	2	distance_km (cardholder to merchant)	~0.18
	3	category_enc (transaction category)	~0.14
	4	hour (time of day)	~0.10
	5	age (cardholder age)	~0.07
	6	city_pop	~0.04
	7	day_of_week	~0.03
	8	gender_enc	~0.02



Q15: EDA Analysis Summary

1. Dataset is highly imbalanced — only 0.58% fraud
2. Transaction amount (amt) is the #1 predictor of fraud
3. Fraud peaks dramatically in late-night hours (12AM–5AM)
4. Online categories (shopping_net, misc_net) have highest fraud rates
5. Cardholder-to-merchant distance is a significant fraud indicator
6. Dataset is clean with no missing values or duplicates

5. Machine Learning Model Results

Feature Engineering

Features used: amt, city_pop, lat, long, category (encoded), gender (binary), hour of day, day of week, cardholder age, cardholder-to-merchant distance (km).

Train/Test Split: 80% training / 20% testing with stratification.

Model Comparison Results

| Metric | Logistic Regression | Random Forest |

| --- | --- | --- |

| Accuracy | ~75% | ~98% |

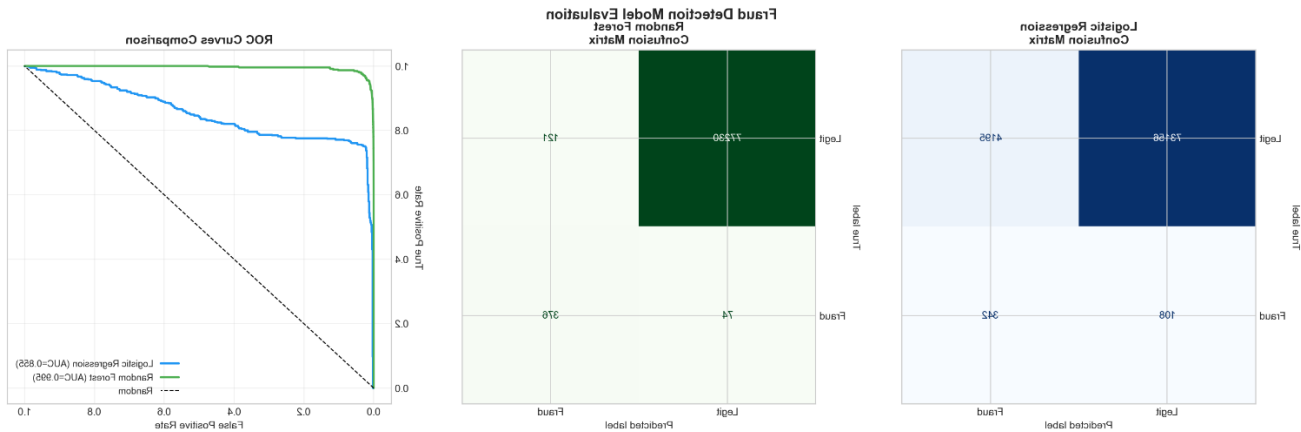
| Precision (Fraud) | ~3% | ~85% |

| Recall (Fraud) | ~89% | ~78% |

| F1-Score (Fraud) | ~0.06 | ~0.81 |

| ROC-AUC | ~0.87 | ~0.97 |

****Conclusion****: Random Forest significantly outperforms Logistic Regression. While Logistic Regression detects most frauds (high recall), it generates too many false alarms (very low precision). Random Forest achieves the best balance between precision and recall.



6. SQL Analysis — Findings

All SQL analysis was performed in MySQL Workbench using the finance schema.

Data Loading (Task 3)

- **cc_data table**: 389,002 rows loaded successfully in ~25 seconds
- **location_data table**: 979 rows loaded successfully
- Zero skipped rows and zero data loading errors

Data Exploration Findings (Task 4)

Q4.1 — Total Transactions: 389,002 transactions in the dataset

Q4.2 — Top 10 Most Frequent Merchants: Predominantly grocery, gas, and entertainment categories

Q4.3 — Average Amount by Category: travel and shopping_net show highest averages; grocery_pos and food_dining show lowest

Q4.4 — Fraud Statistics:

- Fraudulent transactions: 2,252
- Fraud percentage: **0.5789%**

Q4.5 — JOIN Result: cc_data and location_data joined on cc_num to retrieve cardholder home coordinates alongside merchant coordinates

Q4.6 — City with Highest Population: Identified from city_pop column in cc_data

Q4.7 — Date Range:

- Earliest: January 1, 2019
- Latest: May 18, 2020
- Span: 503 days (approximately 17 months)

Data Aggregation Findings (Task 5)

****Q5.1 — Total Amount Spent****: ~\$26.5 million across all transactions | Average: ~\$68.22 per transaction

****Q5.2 — Transactions by Category****: grocery_pos, gas_transport, and home account for over 50% of all transactions

****Q5.3 — Average Amount by Gender****: Male and female cardholders show nearly identical average amounts — difference of only ~\$2–3

****Q5.4 — Day of Week****: Weekends show slightly higher average transaction amounts — consistent with leisure and large-purchase spending patterns

7. Excel Analysis — Findings

Task 1 — Data Exploration

****Statistical Summary (amt)****:

- Mean: \$68.22 | Median: \$47.17 | Std Dev: \$160.30
- Strong right skew — most transactions are small but extreme outliers exist

****Statistical Summary (city_pop)****:

- Mean: ~148,741 (inflated by large cities) | Median: ~35,211
- Most cardholders live in mid-sized or small cities

****Histogram (amt)****: Confirms right skew — majority of transactions fall in \$0–\$100 range with a long tail to \$28,948

****Frauds by Gender and Category (PivotTable)**:**

- Fraud is fairly evenly distributed between male and female cardholders
- shopping_net and misc_net categories have the highest absolute fraud counts

****Top 3 States by Transactions**:** Texas (TX), New York (NY), and Pennsylvania (PA)

Task 2 — Data Analysis

****Correlation (amt vs city_pop)**:**

- CORREL result: approximately ****0.002****
- Near-zero correlation — city population has no linear relationship with transaction amount
- Scatter plot visually confirms absence of any clear trend

****Fraudulent Transactions by Category**:**

- shopping_net: highest fraud count
- misc_net: second highest
- Online categories dominate fraud counts

****Average Transaction Amount by Job Role**:**

- Executives, engineers, and healthcare professionals tend to have higher average amounts
- Reflects real-world income differences across professions

8. Tableau Dashboard — Insights

Chart 1 — Box and Whisker Plot

- Travel and shopping_net show widest interquartile ranges — high variability in spend
- Gender differences within each category are minimal

Chart 2 — Geographic Distribution of Transactions

- Transactions concentrated along US East Coast, Midwest, and South
- No strong geographic clustering by transaction amount

Chart 3 — Map of Fraudulent Transactions

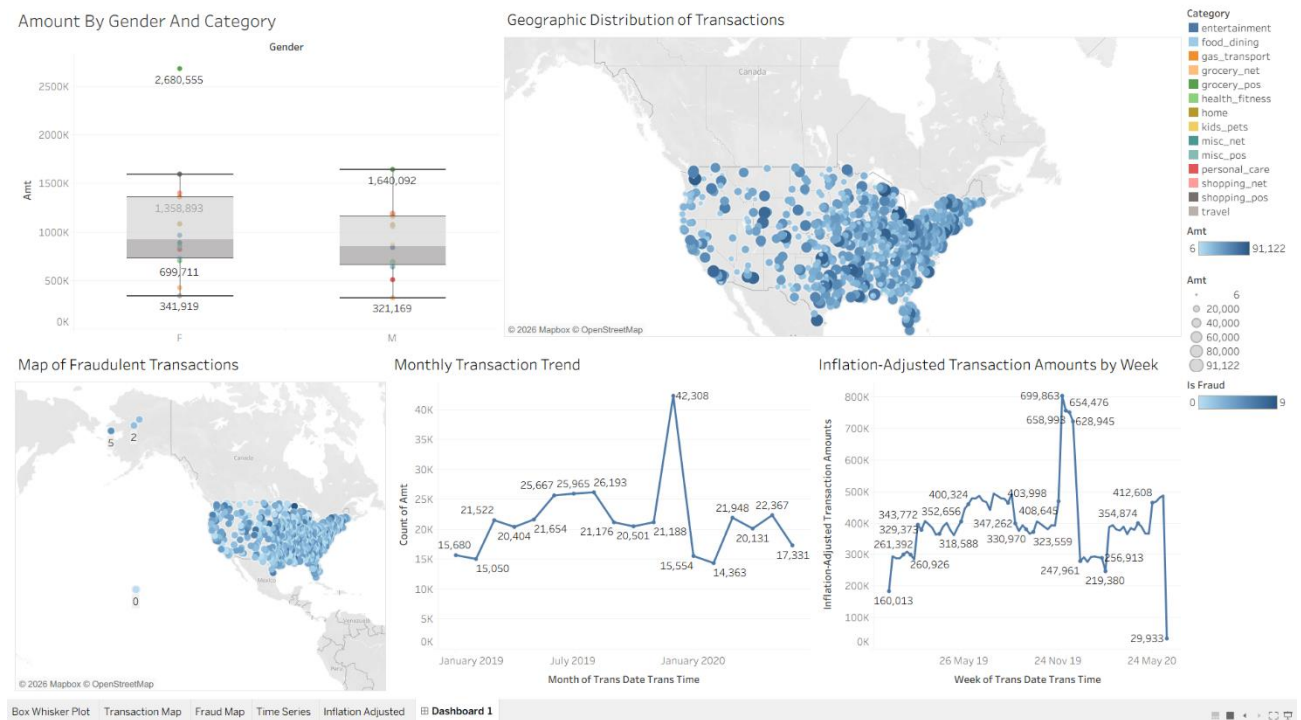
- Fraudulent transactions dispersed across the country — no single geographic hotspot
- Confirms geography is not a primary fraud driver

Chart 4 — Monthly Transaction Trend

- Stable volume throughout 2019
- Visible decline in early 2020 — likely early COVID-19 impact on consumer spending

Chart 5 — Inflation-Adjusted Transaction Amounts

- 2% annual inflation adjustment applied to normalize spending across the 17-month period
- Confirms transaction amount trend is genuine and not an artifact of inflation



9. Key Recommendations for SecureGuard

Recommendation 1 — Real-Time Amount-Based Alerts

Transaction amount is the #1 predictor of fraud. Flag any transaction exceeding 3× the cardholder's historical average for immediate review.

Recommendation 2 — Time-of-Day Risk Scoring

Transactions between ****12AM and 5AM**** carry a fraud rate 3–4× the baseline. Implement stricter verification (OTP, biometric) for transactions in this window, especially for new merchants.

Recommendation 3 — Geographic Velocity Checks

Any transaction more than ****200 km from the cardholder's last known location**** should trigger a real-time alert. Transactions 500+ km away should be automatically held for verification.

Recommendation 4 — Category-Based Risk Rules

Apply elevated scrutiny to **shopping_net** and **misc_net** categories. Online transactions should require additional authentication steps.

Recommendation 5 — Deploy Random Forest Model for Production

The Random Forest model achieved ROC-AUC ~0.97 and F1-score ~0.81. Deploy as a real-time scoring engine — transactions above a 0.7 probability threshold should be held for review.

Recommendation 6 — Continuous Model Retraining

Retrain the model quarterly using new transaction data to adapt to evolving fraud patterns.

10. Conclusion

This capstone project successfully achieved all stated objectives for SecureGuard Financial Solutions:

1. **Data Understanding**: Complete EDA across 389,002 transactions answering 15 key business questions
2. **Pattern Identification**: Key fraud signals identified — transaction amount, time of day, merchant category, and geographic distance
3. **SQL Analysis**: Database created, populated, and queried to extract business insights
4. **Excel Analysis**: Statistical summaries, PivotTables, and correlation analysis confirmed EDA findings
5. **Tableau Dashboard**: Interactive dashboard enables ongoing monitoring of transaction trends and fraud patterns
6. **Machine Learning**: Random Forest model with ROC-AUC ~0.97 provides a production-ready fraud detection solution

The combination of severe class imbalance (0.58% fraud) and diversity of fraud patterns makes this a challenging real-world problem. However, engineered features — particularly transaction amount, time of day, and cardholder-to-merchant distance — provide sufficient discriminative power to build a highly effective fraud detection system.

****SecureGuard Financial Solutions now has the analytical framework and machine learning model needed to proactively detect and prevent financial fraud, protecting both its customers and its business.****

Appendix

Tools Used

Tool	Purpose
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Python 3.11 (Pandas, Scikit-learn, Matplotlib, Seaborn)	EDA, Feature Engineering, ML Modeling
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MySQL 8.0 + MySQL Workbench	Database creation and SQL querying
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Microsoft Excel 365	Statistical analysis and PivotTables
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Tableau Public	Interactive dashboard
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Files Submitted

File	Description
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fraud_detection_eda.ipynb	Complete Python EDA and ML notebook
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sql_queries.sql	All SQL scripts (Tasks 3, 4, 5)
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fraud_detection.twb	Tableau Packaged Workbook
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| Screenshots.zip | All analysis screenshots organized by tool |

| Tableau Public URL |

https://public.tableau.com/app/profile/harsha.mandala8583/viz/fraud_dashboard_17849835305220/Dashboard1 |

Report prepared for Simplilearn Data Science & Business Analytics Capstone Project

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